

*Preface

An inscribed theorem is not a verified theorem. A claim carrying the tag `\ClaimStatusProvedHere` asserts that the reader will find, within the same file, a proof. It does not assert, and cannot by itself assert, that the proof's internal derivation is disjoint from the sources the author consulted to arrive at the statement. The two are routinely identified in mathematical practice. The programme that precedes this volume rests on the Beilinson principle, an explicit adversarial protocol, and over one hundred thousand tests. Yet a trust-based corpus is not a verified corpus.

The deficiency is concrete. Consider the inscription in Vol III of the formula

$$\kappa_{\text{BKM}} = c_N(0)/2$$

for the Borchers–Kac–Moody central charge of a frame-shape orbifold lift. In the compute engine, $c_N(0)$ is tabulated from `FRAME_SHAPE_DATA`. The hardcoded expected values in the test file are computed by halving entries of the same table. Ninety-nine tests pass. The derivation (via the Borchers 1998 weight theorem applied to the input lift) and the verification (by direct arithmetic on the tabulated lift data) share a source. Nothing can fail. No ninety-nine-test suite detects anything about the inscription beyond the arithmetic $c_N(0)/2 = c_N(0)/2$.

This is not a local accident. The source-disjoint audit across the programme found that every programme-wide hardcoded-table verification was of this form: the table from which the theorem was derived also supplied the test expectations. The programme is not verifying that its theorems are true; it is verifying that its arithmetic on a single source is consistent.

The deficiency admits a remedy: a disjointness requirement. A theorem T is verified, in the sense relevant to avoid coincidental-agreement failures of the kind exposed by the $\Psi = 2$ coincidence masking a cross-arity sign error, only if there exists a test whose expected value is computed along a source disjoint from the source used to derive T 's formula. Disjoint means disjoint as a set of citations, not disjoint as a rephrasing. The source-disjoint protocol in Vols I, II, and III (`compute/lib/independent_verification.py`) enforces this: every test decorated with `@independent_verification` must declare its `derived_from` and `verified_against` sources, and the decorator raises `IndependentVerificationError` at collection time if the two sets overlap.

The protocol installs no coverage. The installation snapshot across the three prior volumes covered 2 of 3692 ProvedHere claims: 0/2275 in Vol I, 0/1134 in Vol II, and 2/283 in Vol III. The infrastructure was in place; the content was not. A programme's claim corpus can be verified one decorator at a time, but not by the same act that produces the theorem. The verification must be a separate intellectual operation, carried out with a different source catalog, and held to a different standard than the proof that produced the theorem.

Volume IV is that separate operation.

The realization corpus extends beyond the installation snapshot. Vol I sup-

plies the Virasoro shadow tower with Stokes pole $c_S = -218/45$, Borel radius

$$|\omega|^2(c) = \frac{c^2(5c + 22)}{4(45c + 218)},$$

and Riccati equation $U' + 2U^2 = Q_c''/(2Q_c)$ on $U = (\log H)' - 2/t$; the four-conductor split (K, K^κ, K_c, K_g) ; the $p = 691$ Chenevier determinant; and the eight-form Gritsenko–Cléry spread. Vol II supplies the $\text{SC}^{\text{ch}, \text{top}}$ heptagon and the finite-jet Khan–Zeng (KZ) sector: the shifted-cotangent BV action and the CME/PVA-Hamiltonian equivalence are constructed; the all-loop QME, reflected weight identity, boundary vertex algebra, and E_3 -lift remain conditional on the four endpoint hypotheses, the KZ analytic spectral-decomposition representation (SDR), Fulton–MacPherson Stokes datum, reflected weights, and T -lift. Vol III supplies the two-stage functor

$$\Phi_d = \text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}} \circ \Phi_d^{\text{FA}},$$

the K3 Hall–Drinfeld chiral bialgebra $\mathbf{H}_{\Delta_5}$, the Joyce vertex algebra paired with Kontsevich–Pandharipande–Soibelman (KPS) orientation on generic compact Calabi–Yau threefolds with Maulik–Nekrasov–Okounkov–Pandharipande (MNOP) invariants, S^3 -framing on generic compact Calabi–Yau threefold Lagrangians, and the $so(4, 20)$ K3 Yangian representation-target trichotomy.

The arithmetic branch carries a native Vol IV residual calculus whose central open problem is the spectral-flow question: whether the post-Tate quotient operator admits a self-adjoint extension compatible with the Hadamard residue functional on the critical-line ordinate set Γ_ξ^{crit} . The ordinary M_t -graph dual (the Hilbert space of M_t -measurable functions with the $(1+t^2) dt$ inner product) carries neither point nor finite-jet Hadamard residue functionals on Γ_ξ^{crit} . Any surviving spectral-flow argument requires a stronger topology (rigged Hilbert space, Paley–Wiener, de Branges, reproducing-kernel, nonlocal L^2 , or BV–BFV boundary) in which Hadamard trace and jet maps are continuous and compatible with the post-Tate quotient.

The content is the realization pair

$$\text{Real}(X, T) = (d, e),$$

where d is a source-disjoint decorator whose derivation and verification sources are disjoint, and e is a compute engine whose hardcoded expected values come from a source disjoint from the engine’s formula. The central theorem is that Real exists for every ProvedHere claim inscribed in Vols I, II, III on the non-degenerate locus of the programme. The non-degenerate locus is large: it contains every generic parameter in every standard family, every Koszul-locus claim, every chain-level result outside the admissible logarithmic sector, every CY construction outside the five unconverted CY–C bridges. It excludes the critical level $k = -h^\vee$ for affine Kac–Moody (where $\kappa = 0$ and Koszulness fails), the $\Psi = 0$ locus for chiral Yangians (where the algebra is abelian), the Creutzig–Kanade–Linshaw admissible logarithmic sector, and the super-Yangian chain-level chiral coproduct sector.

Each of the three Vol IV theorems is itself realized: the decorator is installed at inscription time, the engine backs it from a disjoint source catalog, and the reader runs the test on receipt of the repository.

Vol IV inherits the Russian school’s convergent voice: Gelfand’s inevitability (trust-based inscription eventually produces a tautology somewhere in the corpus), Beilinson’s falsification (the source-disjointness check catches the tautology at collection time), Drinfeld’s unifying principle (the realization functor *Real*), Kazhdan’s compression. A fifth voice enters: verification is a craft, and its content is specific. The Borchers $c_N(0)/2$ identification is verified by a Gritsenko–Nikulin denominator identity, not by a tabulation; the affine Kac–Moody $\kappa = \dim(g)(k + h^\vee)/(2h^\vee)$ identification is verified at non-critical level by Feigin–Frenkel screening cohomology, not by character arithmetic on the Sugawara construction; the class M MC5 Wick $S_5 = -48/(c^2(5c + 22))$ is verified by an independent Wick contraction engine built from the Virasoro OPE axioms, not from the lambda-bracket expansion used to derive it.

Structure

Part A indexes every ProvedHere claim from Vols I, II, III; each entry points either to a realized entry in Part B or to a scope-restricted entry with the complementary locus explicitly marked. The realization programme as a mathematical object appears in Chapter ???. The decorator protocol and its collection-time enforcement are implemented in `compute/lib/realization_registry.py`, which extends the Vol I `compute/lib/independent_verification.py`. Conventions are inherited from Vol I and distinguished here by the prefix `v4-` and the Vol IV registry’s import-time disjointness check against the Vol I registry.

Credits

The disjoint source catalogs are cited at the points of use: Pridham–Toen–Vezzosi, Lurie HA 5.3, Costello–Gwilliam, Francis–Gaiatsgory, Mok 2025, Hua–Keller, Ben-Zvi–Francis–Nadler, Schiffmann–Vasserot, Costello–Witten–Yamazaki. The mathematical text is authored by Raez Lorgat.